University Of Asia Pacific

Course Code : CSE 402

Course Title : Operating System Lab

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Submitted to,

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Problem Statement : FCFS & SCAN

Request Queue = [0, 41, 30, 100, 62, 51, 20]

Initial Head Position = 70

Source Code :

1. FCFS Code :

```
request = [0, 41, 30, 100, 62, 51, 20]
head = 70
total_movement = 0

print("Order:", head, end="")

for r in request:
    total_movement += abs(r - head)
    head = r
    print(" ->", head, end="")

print("\nTotal Head Movement =", total_movement)
```

2. Sorted Order Code :

```
request = [0, 41, 30, 100, 62, 51, 20]
head = 70

request.sort()
```

```
total_movement = 0

print("Order:", head, end="")

for r in request:
    total_movement += abs(r - head)
    head = r
    print(" ->", head, end="")

print("\nTotal Head Movement =", total_movement)
```

Output :

```
Order: 70 -> 0 -> 20 -> 30 -> 41 -> 51 -> 62 -> 100
Total Head Movement = 170
```

```
Order: 70 -> 0 -> 41 -> 30 -> 100 -> 62 -> 51 -> 20
Total Head Movement = 272
```

Difference discussion :

The main difference between the two approaches is the **order in which the disk requests are served**.

- In **FCFS**, requests are processed in their original order:
  `70 → 0 → 41 → 30 → 100 → 62 → 51 → 20`
  Therefore, the head moves more, resulting in **272 cylinders** of total head movement.
- In the **Sorted Order approach**, requests are first sorted in ascending order:
  `70 → 0 → 20 → 30 → 41 → 51 → 62 → 100`
  This reduces unnecessary movement and gives **170 cylinders** of total head movement.

So, for this particular input, the **Sorted Order approach performs better than FCFS** because it requires **102 fewer cylinders** of head movement.

GitHub Link : https://github.com/JoynoveMim/Operating-System.git

Discussion :

In this experiment, two different disk request scheduling approaches were implemented using Python: **FCFS and Sorted Order**. FCFS processes requests in the same order as they appear, which causes more disk head movement. In contrast, the Sorted Order approach arranges the requests in ascending order before processing them, which reduces unnecessary movement. For the given request queue, FCFS results in **272 cylinders** of head movement, while Sorted Order results in **170 cylinders**. Therefore, the Sorted Order approach performs better for this particular input. However, Sorted Order is not a standard disk scheduling algorithm like SSTF, SCAN, or C-SCAN.

Conclusion :

In this experiment, **FCFS and Sorted Order disk scheduling approaches** were implemented using Python. The total head movement was calculated for both methods. FCFS resulted in **272 cylinders**, while the Sorted Order approach resulted in **170 cylinders**. Therefore, the Sorted Order approach produced less head movement for the given requests. This experiment helped us understand how the order of servicing disk requests can affect disk performance and access time.